Administrator manual: FaceAttend EDU

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Version Control

Every modification to this manual must be recorded before the document is submitted again. The manual version must always correspond to the version of the software it describes.

Table 1

Document version history

Version	Date	Description of change	Responsible
1.0	06/10/2026	Initial creation of the administrator manual.	Development team
1.1	06/10/2026	Application of APA 7th edition standards and removal of color usage.	Development team
[ ]	[ ]	[Record the next change here.]	[ ]

Source. Own elaboration.

Note. Minor changes, such as wording adjustments or the addition of screenshots, increment the second digit. Functional changes, such as the addition of an administrative module or the modification of permissions, increment the first digit.

Update Rule

- The manual must be reviewed after every deployment that modifies screens, roles, or reports.
- Every delivered version must be recorded in the table above together with its responsible party.

Introduction

FaceAttend EDU is a system built to record and control attendance within a training environment. The system replaces manual attendance taking with a digital workflow, so that the information is centralized, queryable, and exportable.

This manual explains how to administer the system. It does not explain how to use it as a student, nor how the source code is written; that information is found in the user manual and in

the technical manual, respectively. The structure of the document follows the institutional guide for producing software manuals (Servicio Nacional de Aprendizaje, 2026).

The document is written so that a person who did not take part in the development can take over administration of the system by reading it in full. Each procedure indicates where the option is located, what steps to follow, and what result should be produced. The manual contains no passwords, private keys, tokens, or production credentials.

Table 2

Intended readers of the manual and applicable sections

Reader	What they will find in this manual
Super Administrator	The entire document, including the configuration, backup, and restoration sections.
Administrator	Management of users, cohorts, courses, schedules, reports, and audit.
Instructor with delegated functions	Sections 9, 12, and 13 only, and only for the groups assigned to them.
Person taking over the project handover	The complete document, starting with sections 7 and 8.

Source. Own elaboration.

Table 3

Complementary project documents

Document	Question it answers	When to consult it
User manual	How do I use the system?	When an end user asks about a screen or a message.
Technical manual	How is the system built?	When services, the database, or endpoints need to be understood.
Installation manual	How do I install and run the system?	When the system is being brought up on a new machine or server.
Administrator manual	How do I administer the system?	This document. Operation of the already-installed system.

Source. Own elaboration.

Objective

To guide the administrator in the management, configuration, and supervision of FaceAttend EDU, ensuring secure and organized use of the platform.

Specific Objectives

1. Describe the profile, responsibilities, and limits of the administrator role within the system.
2. Document the procedure for accessing the administration panel.
3. Establish the procedures for creating, modifying, deactivating, and resetting user accounts.
4. Define the existing roles and the permissions associated with each one.
5. Explain the administration of cohorts, courses, and schedules.
6. Document the generation, querying, and export of attendance reports.
7. Establish the backup and restoration procedure for the system's two databases.
8. Define the security practices the administrator must comply with.

Scope

Content Covered by the Manual

- Administration of user accounts and role assignment.
- Administration of cohorts, courses, and schedules.
- Querying, filtering, and exporting attendance reports.
- General system configuration parameters.
- Backup and restoration of PostgreSQL and MongoDB.
- Querying the history of administrative actions.
- Incident handling and escalation procedures.

Content Not Covered by the Manual

- Installation, deployment, and initial configuration of the microservices, described in the installation manual.
- Internal architecture, detailed data model, and endpoints, described in the technical manual.
- Day-to-day use of the system by students and instructors, described in the user manual.

- Administration of the server or container infrastructure where the system runs.
- Modification of the source code.

Every procedure in this manual is carried out on the already-installed and running system. If the system is not installed, the installation manual must be applied first.

Administrator Profile

The administrator is the person responsible for keeping the system available, for ensuring that each user has the access that corresponds to them, and for ensuring that the attendance information is reliable.

Table 4

Profile required for the administrator role

Aspect	Description
Knowledge of the system	Must be familiar with the users, cohorts, courses, schedules, and reports modules.
Minimum technical knowledge	Use of a web browser, downloading and storing files, and basic notions of databases in order to interpret a backup.
Institutional judgment	Must know the structure of cohorts, programs, and shifts of the training center.
Responsibility over data	Handles personal data of students and instructors, and is therefore accountable for its confidentiality.

Source. Own elaboration.

Administrator Responsibilities

Daily Responsibilities

- Verify that the system is responding and that attendance recording is operating.
- Handle access requests: account creation, password reset, and unlocking.
- Review the issues reported by instructors regarding unrecorded attendance.

Weekly Responsibilities

- Review the list of active users and deactivate accounts that no longer correspond to people involved in the process.
- Generate and archive the attendance report for the period.
- Verify that the week's backup was generated and stored correctly.

Monthly Responsibilities

- Review the history of administrative actions, looking for unjustified operations.
- Validate that the system configuration still corresponds to the reality of the training center.
- Verify that this manual still matches the actual screens and procedures of the system.

Event-Driven Responsibilities

- Run a backup before any major change or deployment.
- Coordinate the restoration of information when data loss occurs.
- Escalate to the technical team those incidents that cannot be resolved from the administration panel.

Limits of the Role

The administrator must not perform the following actions:

- Directly modify attendance records in order to alter an academic result.
- Share their account with another person or use a generic shared account.
- Delete users who have attendance history. Instead, the account is deactivated, so that the history is preserved.
- Modify the source code or the database directly without authorization from the technical lead.

Access to the Administration Panel

The administration panel is the interface from which all the procedures in this manual are carried out. Access depends on the role assigned to the account.

Table 5

Prerequisites for accessing the administration panel

Requirement	Detail
Browser	Google Chrome, Microsoft Edge, or Mozilla Firefox, in an up-to-date version.
Connection	Access to the network where the system is published.
Account	User with the Super Administrator or Administrator role, in Active status.

Requirement	Detail
System address	http://localhost:8090
Services running	The system microservices and the PostgreSQL and MongoDB databases must be available.

Source. Own elaboration.

Login Procedure

1. Open the browser and enter the system address.
2. On the login screen, type the institutional email address and the password.
3. Click the Log in button.
4. Verify that the system displays the panel corresponding to the administrator role.
5. Confirm that the administration menu is visible.

Expected result: the system presents the administration panel with the user's name and role at the top.

The screenshots corresponding to this procedure are included in Appendix E.

Table 6

Frequent access problems and corrective actions

Situation	Probable cause	Administrator action
The system reports invalid credentials	Incorrect email or password.	Verify the email address. If it persists, reset the password as described in section 9.
The system reports that the account is inactive	The account was deactivated.	Reactivate the account if appropriate, as described in section 9.
The user logs in but does not see the admin menu	The assigned role is not an administrative one.	Check the account's role in section 10 and correct it if appropriate.
The page does not load	Service stopped or incorrect address.	Verify the availability of the services and escalate to the technical team as per section 17.
The session closes by itself	The configured inactivity timeout elapsed.	Log in again and review the session timeout parameter in section 11.

Source. Own elaboration.

User Management

The users module allows accounts in the system to be created, queried, edited, activated, deactivated, and have their password reset. It is the module most frequently used by the administrator.

Table 7

Fields that make up a user account

Field	Mandatory	Remark
First and last names	Yes	Must match the person's identification document.
Identification number	Yes	Unique in the system. It cannot be registered twice.
Email	Yes	This is the credential the person logs in with. It must be unique.
Phone	No	Used for contact in the event of an issue.
Role	Yes	Defines what the person can do. See section 10.
Status	Yes	Active or Inactive. Determines whether the person can log in.
Associated cohort or group	Depends on role	Applies to students and instructors.

Source. Own elaboration.

Create a User

1. Log in with an administrator account.
2. Select the Administration menu and then the Users option.
3. Click the New user button.
4. Fill in first names, last names, identification number, email, and phone.
5. Select the role that corresponds to the person's actual function.
6. Associate the cohort or group when the role requires it.
7. Set the status to Active.
8. Click Save.
9. Verify that the user appears in the list and that the system displays the successful registration message.

Expected result: the user is created and can access the system according to the assigned role.

Before creating an account, the identification number must be searched for in the list. If the person already exists with Inactive status, the existing account must be reactivated instead of creating a new one, since a duplicate record splits that person's attendance history.

Edit a User

1. Go to the Administration menu and then to Users.
2. Search for the person by name, identification number, or email.
3. Click the Edit option on the corresponding row.
4. Modify only the fields that need correction.
5. Click Save and confirm the successful update message.

Role changes and status changes are recorded in the action history described in section 16.

Activate or Deactivate a User

Deactivation is the standard procedure when a person stops being part of the training process, since it preserves the attendance history and blocks access.

1. Go to the user list.
2. Search for the person.
3. Change the status to Inactive.
4. Save the change.
5. Verify that the account no longer allows logging in.

Table 8

Effects of activating, reactivating, and deleting accounts

Action	When it applies	Effect on the history
Deactivate	The person completed their process, withdrew, or changed role.	The attendance history is preserved intact.
Reactivate	The person returns to the training process.	The previous history of the same account is recovered.

Action	When it applies	Effect on the history
Delete	Only for records created in error and with no associated attendance.	The record is lost. Requires Super Administrator authorization.

Source. Own elaboration.

Reset a User's Password

1. Go to the user list and search for the person.
2. Select the Reset password option.
3. Confirm the action.
4. Inform the person, through a verifiable channel, that they must set a new password on their first login.

The administrator must never know, write down, or store another user's final password. The reset forces the user to define their own password.

Query and Filter Users

- Filter by role to review how many administrative accounts exist.
- Filter by status to identify active accounts that should no longer be active.
- Filter by cohort or group to validate the composition of a group before a period starts.
- Search by identification number or email to handle a specific support request.

Role and Permission Management

FaceAttend EDU handles four roles. Each role determines which modules the person sees and which actions they can carry out. The assignment principle is to grant only the permissions that the person's actual function requires.

Table 9

System roles, main permissions, and restrictions

Role	Main permissions	Restrictions
Super Admin	Full control of the system. Manages administrators, general configuration, backups, and restoration. Accesses the complete action history.	A minimum number of accounts with this role must exist. It is not shared between people.

Role	Main permissions	Restrictions
Administrator	Manages users, cohorts, courses, schedules, and reports. Queries the action history.	Does not perform database restorations or modify critical parameters without Super Admin authorization.
Instructor	Queries and manages attendance for the cohorts and courses assigned to them. Generates reports for their groups.	Does not create or delete users. Does not access information for groups not assigned to them.
Student	Queries their own attendance and personal information.	Does not access other users' information. Does not generate administrative reports.

Source. Own elaboration.

Assign or Change a Role

1. Go to the Administration menu and then to Users.
2. Search for the person and select Edit.
3. Change the Role field to the appropriate value.
4. Save the change.
5. Ask the person to log out and log back in so that the permissions take effect.
6. Confirm with the person that they see the expected modules.

A role change immediately modifies what the person can see and do. Before elevating a role to Administrator or Super Admin, an explicit request from the project owner must exist. This change is recorded in the action history.

Permission Matrix by Operation

The following matrix summarizes who can carry out each operation. It must be validated against the actual behavior of the system and adjusted if that behavior changes.

Table 10

Permission matrix by operation and role

Operation	Super Admin	Administrator	Instructor	Student
Create user	Yes	Yes	No	No
Change a user's role	Yes	Yes, except Super Admin	No	No
Deactivate user	Yes	Yes	No	No
Create cohort or course	Yes	Yes	No	No

Operation	Super Admin	Administrator	Instructor	Student
Assign schedule	Yes	Yes	No	No
Record or adjust attendance	Yes	Yes	Own groups only	No
Generate attendance report	Yes	Yes	Own groups only	Own only
Export report	Yes	Yes	Own groups only	No
Modify system configuration	Yes	Partial	No	No
Generate backup	Yes	Subject to authorization	No	No
Restore information	Yes	No	No	No
Query action history	Yes	Yes	No	No

Source. Own elaboration.

Note. The value Partial indicates that the role has access to some parameters and not to others. The detail is specified in section 11.

System Configuration

The configuration defines the general parameters the system operates with. A change in this section affects all users, so every modification must be recorded and communicated.

Table 11

Administrable system parameters

Parameter	Description	Reference value
Institution name	Data that appears in headers and reports.	Centro de la Industria, la Empresa y los Servicios — SENA
Session timeout	Minutes of inactivity before the session is closed automatically.	30 minutes
Attendance statuses	Values allowed when recording attendance.	Present, Absent, Late, Excused
Check-in tolerance	Minutes after the start time within which the record is still considered on time.	Defined by the training center
Shifts	Time bands in which training operates.	Morning, Afternoon, Evening
Export formats	Formats available for downloading reports.	PDF and Excel
Current academic period	Date range over which reports are consolidated.	Defined by the training center

Source. Own elaboration.

Procedure for Modifying a Parameter

1. Generate a backup before the change, as described in section 14.
2. Go to the Administration menu and then to Configuration.
3. Locate the parameter that needs to be modified.
4. Record the previous value in a note, in case it has to be reverted.
5. Apply the new value and save.
6. Verify the effect of the change with a concrete test.
7. Inform the affected users as described in section 17.

Parameters External to the Panel

The following elements belong to the deployment configuration and are not modified from the administration panel. Any change to them is the responsibility of the technical team and is documented in the installation manual and in the technical manual.

Figure 1

Structure of the system environment variables

```
POSTGRES_HOST=...
POSTGRES_PORT=...
POSTGRES_DB=...
POSTGRES_USER=...
POSTGRES_PASSWORD=YOUR_VALUE
MONGO_URI=...
MONGO_DB=...
EMAIL_INTERNAL_TOKEN=YOUR_VALUE
SESSION_TIMEOUT=...
```

Source. Own elaboration.

Note. The figure presents only the structure of the variables. The manual must not publish passwords, private keys, tokens, or real connection strings. There is no JWT secret: sessions are

opaque UUID identifiers stored in the identity database, so there are no signing keys to configure or rotate.

Management of Core Information

The core information of FaceAttend EDU is made up of cohorts, courses, and schedules. Correct configuration of these determines whether attendance recording is associated with the right group and the right session.

Figure 2

Configuration order of the core information

1. Cohort (training group)

|

2. Course (associated with a cohort)

|

3. Schedule (associated with a course and an instructor)

|

4. Students (associated with the cohort)

|

5. Attendance recording (generated against the schedule)

Source. Own elaboration.

Note. The elements have dependencies among themselves. Configuring them in a different order produces assignment errors.

Manage Cohorts

Create a Cohort

1. Go to the Administration menu and then to Cohorts.
2. Click New cohort.
3. Fill in the cohort number, program name, shift, and start and end dates.
4. Save and verify that the cohort appears in the list.

Cohort Administration Rules

- The cohort number is unique and must not be registered twice.
- A cohort with recorded attendance is not deleted; it is closed or marked as finished.
- When a cohort is closed, verify that the reports for the period have already been generated and archived.

Manage Courses

1. Go to the Administration menu and then to Courses.
2. Click New course.
3. Fill in the name of the course or competency and the description.
4. Associate the course with the corresponding cohort.
5. Assign the responsible instructor.
6. Save and verify the record in the list.

Expected result: the course becomes available for schedule assignment and the assigned instructor sees it on their panel.

Manage Schedules

1. Go to the Administration menu and then to Schedules.
2. Click New schedule.
3. Select the course.
4. Define the days of the week, the start time, and the end time.
5. Confirm the environment or classroom when the system requests it.
6. Save and verify that there is no overlap with another schedule of the same instructor or the same group.

Table 12

Validations applicable to schedule assignment

Validation	What to check	Action if it fails
Group schedule overlap	That the cohort does not have two simultaneous sessions.	Adjust the time band of one of the sessions.

Validation	What to check	Action if it fails
Instructor schedule overlap	That the instructor is not assigned to two groups at the same time.	Reassign the instructor or change the time band.
Consistency with the shift	That the time band corresponds to the cohort's shift.	Correct the shift or the time band.
Validity period	That the schedule falls within the cohort's dates.	Adjust the dates before saving.

Source. Own elaboration.

Attendance Issue Management

When an instructor or a student reports that an attendance record was not registered or was registered incorrectly, the administrator applies the following procedure.

1. Receive the request through the official channel, including the date, group, student name, and description of the issue.
2. Look up the record in the corresponding attendance report.
3. Verify the issue with the instructor responsible for the session.
4. Apply the adjustment in the system only if the issue is confirmed.
5. Record in the issue log the date, the responsible party, and the justification for the adjustment.
6. Inform the requester that the issue has been handled.

Every manual attendance adjustment must be justified and recorded. An unsupported adjustment compromises the reliability of the system and of the administrator.

Modules Pending Documentation

If the system implements face enrollment, attendance recording by facial recognition, or other additional administrative modules, the team must document them in this section using the same format employed in the previous sections: numbered steps, expected result, validations, and a real screenshot. Modules that are not implemented must not be documented.

Administrative Reports

Reports are the final product of the system. They make it possible to verify attendance by group, by person, and by period, and they support academic decisions.

Table 13

Reports available in the system

Report	Content	Filters	Frequency
Attendance by cohort	Consolidated attendance of all the students in a group.	Cohort and date range.	Weekly
Attendance by student	Individual attendance history.	Student and date range.	On request
Attendance by course	Attendance grouped by course or competency.	Course and date range.	Weekly
Attendance by instructor	Sessions and records associated with an instructor.	Instructor and date range.	Monthly
Registered users	System accounts by status and by role.	Role and status.	Monthly
Accumulated absences	Students who exceed the absence threshold.	Cohort and period.	Weekly

Source. Own elaboration.

Generate and Export a Report

1. Go to the Reports menu.
2. Select the report type.
3. Apply the cohort, course, student, or instructor filters and the date range.
4. Click Generate or Query.
5. Check on screen that the data corresponds to the applied filter.
6. Select the export format, PDF or Excel.
7. Download the file and save it using the defined naming convention.

Expected result: the system generates the file with the filtered data and downloads it to the machine.

Figure 3

Naming convention for report files

Report_<type>_<identifier>_<YYYY-MM-DD>.pdf

Examples:

Report_attendance_cohort_<number>_2026-10-06.pdf

Report_active_users_2026-10-06.xlsx

Source. Own elaboration.

Monitoring Metrics

The administrator monitors the indicators in the following table. The target values must be set by the team according to the agreements of the training center.

Table 14

Administrative monitoring indicators

Indicator	Source	Periodicity	Alert threshold
Attendance percentage by cohort	Attendance by cohort report.	Weekly	Defined by the training center
Students with critical absence	Accumulated absences report.	Weekly	Defined by the training center
Sessions with no attendance record	Comparison between scheduled sessions and generated records.	Weekly	Any session without a record
Active accounts not in use	User report cross-referenced with the last login.	Monthly	Defined by the training center
Attendance issues handled	Issue log.	Monthly	More than 48 hours unattended
Backups generated	Backup folder.	Weekly	Any week without a backup

Source. Own elaboration.

Good Practices in Report Handling

- Always verify the date range before delivering a report, since an incorrect filter produces incorrect conclusions.

- Archive the period's reports in a controlled folder and not on the desktop of a personal machine.
- Deliver reports only to those whose function justifies it, as they contain students' personal data.
- Do not manually modify the exported file. If a data point is wrong, it is corrected in the system and the report is generated again.

Backups

FaceAttend EDU uses two databases. A complete backup must include both, since backing up only one leaves the system in an inconsistent state at the time of restoration.

Table 15

Elements included in the backup

Element	Engine	Content	Criticality
Relational database	PostgreSQL	Users, roles, cohorts, courses, schedules, and attendance records.	High
Document database	MongoDB	Unstructured system information, according to the service that manages it.	High
Configuration files	Deployment files	Microservice parameters, without real credentials.	Medium
Archived reports	File system	Exported reports from closed periods.	Medium

Source. Own elaboration.

Table 16

Backup frequency

Moment	Scope	Responsible
Weekly	Complete backup of PostgreSQL and MongoDB.	Administrator
Before a deployment or update	Complete backup.	Super Admin or technical lead
Before a major configuration change	Complete backup.	Administrator
At the close of an academic period	Complete backup plus the period's report archive.	Administrator

Source. Own elaboration.

Procedure from the Panel

1. Log in to the administration panel with an authorized account.
2. Select the Backups option.
3. Click Generate backup.
4. Wait for the system to confirm generation.
5. Download the generated file.
6. Save it in the backup folder with the date in the file name.
7. Record the backup in the backup log.

Figure 4

Naming convention for backup files

backup_faceattend_postgres_2026-10-06.sql

backup_faceattend_mongo_2026-10-06.archive

Source. Own elaboration.

Alternate Command-Line Procedure

When the panel does not offer the option or is unavailable, the backup is run by the technical team using the native tools of each engine. The connection values are taken from the deployment environment variables.

Figure 5

Database backup commands

```
# PostgreSQL
pg_dump -h <host> -p <port> -U <user> -d <database> \
-F c -f backup_faceattend_postgres_<YYYY-MM-DD>.dump
# MongoDB
mongodump --uri="<connection_string>" \
--archive=backup_faceattend_mongo_<YYYY-MM-DD>.archive --gzip
```

Source. Own elaboration.

Note. The commands are run with credentials that must not be written in this manual nor shared over open channels. They must be requested from the technical lead at the time of use.

Backup Storage

- Store each backup in at least two different locations.
- Keep one of the locations off the machine where the system runs.
- Retain at minimum the four most recent weekly backups and the closing backup of each period.
- Restrict access to the backup folder, since it contains complete personal data.
- Verify that the downloaded file has a plausible size, as a file of a few kilobytes indicates a failed backup.

Table 17

Backup log

Date	Type	Files generated	Location	Responsible	Verified
[ ]	[Weekly, pre-deployment, or closing]	[ ]	[ ]	[ ]	[Yes or No]
[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
[ ]	[ ]	[ ]	[ ]	[ ]	[ ]

Source. Own elaboration.

Information Restoration

Restoration returns the system to the state of the selected backup. It is a procedure that overwrites information, so it is carried out only by the Super Administrator or the technical lead, and always with prior authorization.

All information recorded between the backup date and the moment of restoration is lost. Before restoring, a backup of the current state must be generated, so that it is possible to roll back if the procedure fails.

Table 18

Criteria for determining the need for restoration

Situation	Does it require restoration?	Prior action
Massive loss of attendance records	Yes	Confirm the scope with the technical team.
A user deleted by mistake	Not necessarily	Recreate the user and restore only if their history was lost.
Error in a single attendance record	No	Correct it from the corresponding module, as per section 12.
Corrupt or inaccessible database	Yes	Escalate to the technical lead before taking any action.
Failed deployment that altered the data	Yes	Restore the backup taken before the deployment.

Source. Own elaboration.

Restoration Procedure

1. Obtain authorization from the project owner and have it in writing.
2. Inform users that the system will be unavailable during the procedure.
3. Generate a backup of the current state of both databases.
4. Stop the services that write to the databases.
5. Select the backup file corresponding to the required date.
6. Restore PostgreSQL and MongoDB to the same point in time.
7. Restart the services.

8. Verify the system according to the checklist in the following section.
9. Inform users that the system has been restored and state the date up to which the information is valid.
10. Record the event in the incident log.

Post-Restoration Verification

- Login works with an administrator account.
- The user list shows the expected number of records.
- Roles and permissions are correctly preserved.
- Cohorts, courses, and schedules appear complete.
- An attendance report for a known period yields the expected data.
- The most recent records correspond to the date of the restored backup.

Alternate Command-Line Procedure

Figure 6

Database restoration commands

```
# PostgreSQL
pg_restore -h <host> -p <port> -U <user> -d <database> \
--clean --if-exists backup_faceattend_postgres_<YYYY-MM-DD>.dump
# MongoDB
mongorestore --uri="<connection_string>" \
--archive=backup_faceattend_mongo_<YYYY-MM-DD>.archive --gzip --drop
```

Source. Own elaboration.

Audit and Action History

The action history makes it possible to establish who carried out each operation and when. It is the tool with which the administrator substantiates any change to users, permissions, or attendance records.

Table 19

Actions that must be recorded in the history

Action	Minimum expected data	Who reviews it
User creation	User created, role assigned, responsible party, date and time.	Administrator
Role change	Previous role, new role, responsible party, date and time.	Super Admin
Account activation or deactivation	Previous status, new status, responsible party, and date.	Administrator
Password reset	Account affected, responsible party, and date.	Super Admin
Adjustment of an attendance record	Record affected, previous value, new value, and responsible party.	Super Admin
Configuration change	Parameter, previous value, new value, and responsible party.	Super Admin
Backup generation or restoration	Type of operation, responsible party, and date.	Super Admin

Source. Own elaboration.

Query the History

1. Go to the Administration menu and then to Audit or Action history.
2. Apply the date range filter.
3. Filter by responsible user or by action type when investigating a specific case.
4. Review the resulting records.
5. Export the result when the review must be kept as evidence.

Periodic Review

Once a month the administrator reviews the history, looking specifically for the following items:

- Role changes to Administrator or Super Admin that do not correspond to a recorded request.
- Attendance adjustments with no justification in the issue log.
- Administrative access at unusual hours.
- Repeated password resets on the same account.

- Deletion of records.

Any finding is documented and escalated according to the procedure in section 17.

Actions Not Logged by the System

If the current system does not log one of the indicated actions, this must be noted in this section and reported to the technical team as a pending improvement. In the meantime, that action is recorded manually in the administrative change log in Appendix C.

Security Recommendations and Support Procedures

Account Security

- Do not share the administrator password with anyone, through any channel.
- Use individual accounts, with no generic accounts shared among several people.
- Assign permissions according to each person's actual function and not out of convenience.
- Immediately deactivate the accounts of people who no longer belong to the process.
- Log out when finished using the system, especially on shared machines.
- Do not leave the administrator session open on an unattended machine.
- Change your own password at any suspicion that it has become known to another person.

Information Security

- Take a backup before any major change.
- Do not publish credentials, private keys, tokens, or connection strings in documents, repositories, or messages.
- Treat reports as personal information and deliver them only to those whose function justifies it.
- Do not store backups or reports on unauthorized personal services.
- Review the action history periodically.

Communication with Users

Table 20

User communication matrix

Situation	Recipient	Timing	Channel
Scheduled maintenance	All users	48 hours in advance	Institutional email
Unplanned system outage	Instructors and coordination	As soon as it is detected	Institutional email
Change in roles or permissions	Affected user and their supervisor	Before applying the change	Institutional email
Information restoration	All users	Before and after the procedure	Institutional email
New system version	All users	On the day of deployment	Institutional email
Attendance issue handled	Whoever reported the issue	On closing the case	Institutional email

Source. Own elaboration.

Incident Management and Escalation

An incident is any situation that prevents normal use of the system or that compromises the information. The procedure is as follows.

1. Record the incident with date, time, reporting user, description, and a screenshot of the error.
2. Classify the level according to the following table.
3. Attempt resolution from the administration panel when the level allows it.
4. Escalate to the corresponding level if it is not resolved within the defined time.
5. Inform the reporting user, both on receiving the case and on closing it.
6. Record the applied solution in the incident log in Appendix C.

Table 21

Incident escalation levels

Level	Examples	Responsible	Response time
1. Operational	Forgotten password, inactive account, or a question about a report.	Administrator	Same day
2. Functional	A module does not respond, a report yields incorrect data, or a schedule overlap occurs.	Administrator with support from the development team	1 week
3. Technical	A microservice down, a connection error to PostgreSQL or MongoDB, or an inaccessible system.	Technical lead	Immediate
4. Critical	Information loss, unauthorized access, or database corruption.	Super Admin and technical lead	Immediate, with notification to the project owner

Source. Own elaboration.

Change Management

A change is any planned modification to the system, such as a new version, a configuration change, the modification of permissions, or the adjustment of parameters. The procedure is as follows.

1. Request: whoever requires the change submits it in writing, stating the reason.
2. Assessment: the administrator and the technical lead determine the impact and the affected users.
3. Authorization: the project owner approves or rejects the change.
4. Backup: a backup is generated before applying the change.
5. Application: the change is carried out in the agreed window.
6. Verification: it is checked that the system works and that the change produced the expected effect.
7. Communication: the affected users are informed.
8. Documentation: this manual and its version control are updated when the change modifies a procedure.

If an applied change produces unexpected behavior, it is rolled back using the previous backup before attempting successive corrections, since accumulating corrections on an unstable system makes it harder to identify the cause.

Known Risks and Contingency Plan

Table 22

Known risks and contingency plan

Risk	Impact	Prevention	Contingency
One of the microservices goes down	One module stops working while the rest operates.	Service monitoring and daily review.	Escalate to level 3 and record attendance on paper until service is restored.
Loss of connection to PostgreSQL	The system does not allow information to be queried or recorded.	Weekly backup and availability checks.	Escalate to level 3 and restore from the latest backup if there is corruption.
Loss of connection to MongoDB	The information managed by the document service fails.	Weekly backup at the same point in time as PostgreSQL.	Escalate to level 3 and restore both databases to the same point.
Unverified backup that fails on restore	Permanent loss of information.	Verify the file size and test a trial restoration once per period.	Use the immediately preceding backup and document the extent of the loss.
Compromised administrative account	Unauthorized access to personal data.	Individual accounts, logging out, and review of the history.	Deactivate the account, reset credentials, review the history, and escalate to level 4.
Loss of the only active administrator	Nobody can manage the system.	Keep at least two active administrative accounts.	The Super Admin creates a new administrative account.
Drift between the manual and the system	Procedures that do not match reality.	Review of the manual after every deployment.	Update the manual and record the new version in section 2.

Source. Own elaboration.

Glossary

Table 23

Glossary of system terms

Term	Definition
Administrator	Role with permissions to manage users, cohorts, courses, schedules, and system reports.
Student	Person in a training process whose attendance the system records.

Term	Definition
Audit	Log of the actions carried out in the system, with responsible party, date, and detail of the change.
Backup	File containing a copy of the system's information at a given point in time.
Course	Training unit associated with a cohort, against which schedules are planned.
Escalation	Procedure for transferring an incident to a higher level of support.
Account status	Condition that determines whether a user can log in: Active or Inactive.
Export	Download of a report's information in PDF or Excel format.
Cohort	Training group identified by a unique number, to which the students belong.
Schedule	Band of days and hours in which a course takes place and against which attendance is recorded.
Incident	Situation that prevents normal use of the system or compromises the information.
Instructor	Role that manages attendance for the cohorts and courses assigned to them.
Microservice	Independent system component that provides a specific function and communicates with the others.
MongoDB	Document database engine used by the system.
Attendance issue	Request for correction of an attendance record.
Permission	Authorization to carry out a given action within the system.
PostgreSQL	Relational database engine used by the system.
Restoration	Procedure that returns the system to the state contained in a backup.
Role	Set of permissions that defines what a user can do within the system.
Super Admin	Role with full control of the system, including general configuration and information restoration.
Environment variable	Configuration parameter external to the code with which the services are run.

Source. Own elaboration.

References

Servicio Nacional de Aprendizaje. (2026). *Guía para elaborar manuales de software: Manual de usuario, manual técnico, manual de instalación, manual de administrador* [Support material for students]. Centro de la Industria, la Empresa y los Servicios.

Appendix A

Administrator Activity Checklist

Table 24

Administrator activity checklist

Activity	Periodicity	Complies	Date	Remark
The system responds and allows login	Daily	[Yes or No]	[ ]	[ ]
Access requests handled	Daily	[Yes or No]	[ ]	[ ]
Attendance issues handled	Daily	[Yes or No]	[ ]	[ ]
Active users reviewed	Weekly	[Yes or No]	[ ]	[ ]
Attendance report generated and archived	Weekly	[Yes or No]	[ ]	[ ]
Backup generated and verified	Weekly	[Yes or No]	[ ]	[ ]
Action history reviewed	Monthly	[Yes or No]	[ ]	[ ]
System configuration validated	Monthly	[Yes or No]	[ ]	[ ]
Manual checked against the real system	Monthly	[Yes or No]	[ ]	[ ]

Source. Own elaboration.

Appendix B

Contact Directory

The team must fill in this directory before delivering the manual. Without it, the escalation procedure in section 17 is not applicable.

Table 25

Project contact directory

Function	Name	Email	Reason for contact
Super Administrator	[TO BE COMPLETED]	[TO BE COMPLETED]	Restorations, critical configuration changes, and level 4 incidents.
Principal administrator	[TO BE COMPLETED]	[TO BE COMPLETED]	Daily management of users, cohorts, schedules, and reports.
Backup administrator	[TO BE COMPLETED]	[TO BE COMPLETED]	Cover for the principal administrator.
Technical lead	[TO BE COMPLETED]	[TO BE COMPLETED]	Level 3 incidents, deployments, and service or database failures.
Development team	[TO BE COMPLETED]	[TO BE COMPLETED]	Functional errors and improvement requests.
Project instructor	[TO BE COMPLETED]	[TO BE COMPLETED]	Academic decisions and approval of changes.

Function	Name	Email	Reason for contact
Project owner	[TO BE COMPLETED]	[TO BE COMPLETED]	Authorization of changes and of restorations.

Source. Own elaboration.

Appendix C

Administrative Change and Incident Log

Table 26

Administrative change and incident log

Date	Type	Description	Responsible	Outcome
[ ]	[Change, incident, or issue]	[ ]	[ ]	[ ]
[ ]	[ ]	[ ]	[ ]	[ ]
[ ]	[ ]	[ ]	[ ]	[ ]
[ ]	[ ]	[ ]	[ ]	[ ]

Source. Own elaboration.

Appendix D

Administrative Activity Calendar

Table 27

Administrative activity calendar

Activity	Frequency	Suggested timing	Responsible
System availability check	Daily	Start of the working day	Administrator
Handling of access requests	Daily	During the working day	Administrator
Generation of the attendance report	Weekly	Friday	Administrator
Backup	Weekly	Friday, at close of day	Administrator
Cleanup of inactive users	Weekly	Friday	Administrator
Review of the action history	Monthly	Last business day of the month	Administrator
Validation of the configuration	Monthly	Last business day of the month	Super Admin
Restoration test	Per period	Before the close of the period	Technical lead
Manual update	Per deployment	After every version	Development team
Academic period closing	Per period	Defined by the training center	Administrator

Source. Own elaboration.

Appendix E

System Screenshots

This appendix must contain the real screenshots of the system, at good resolution and ordered according to the administration flow: login, administration panel, user list, new user form, role management, cohorts, courses, schedules, reports screen, and backups screen.

Each screenshot must be inserted using the figure format established by the APA standards: the label Figure followed by the corresponding number in bold, the title in italics on the following line, the image, and an explanatory note where necessary. The screenshots must not show real personal data or credentials.

Appendix F

Items Pending Completion

Before the final submission, the team must replace all fields marked `[TO BE COMPLETED]` and verify the following points:

- Cover page data: team members, training center, cohort, and instructor.
- URL of the administration panel.
- Real names of the menus and buttons, if they differ from those used in this manual.
- Values of the configuration parameters and thresholds of the metrics.
- Official communication channels and response times.
- Contact directory in Appendix B.
- Real screenshots in Appendix E.
- Authorship details of the institutional guide in the reference list.
- Documentation of the additional implemented modules, as per section 12.